

COMS 4720/5720: Homework 2
Due: February 13th, 11:59pm
Total Points: 40

Late submission policy. An assignment that is submitted one day late will get a penalty of 10%. An assignment that is submitted two days late will get a penalty of 20%. Assignments submitted after two days will not be graded and will get no points. For example, if an assignment is due on Friday by midnight, a submission on Saturday will be penalized by 10%, and a submission on Sunday will be penalized by 20%. A submission on Monday will get no points.

Submission format. Homework solutions must be **handwritten**. You may either write directly on a tablet or write on paper and scan your work. In both cases, the final submission must be compiled into a **single PDF file**. Typed solutions are not permitted unless you obtain prior, explicit permission from the instructor. We reserve the right **NOT** to grade homework that does not follow these requirements.

The submission file must be named as follows:

- <Your-net-id>-4720-hw2.pdf for COMS 4720
- <Your-net-id>-5720-hw2.pdf for COMS 5720.

For instance, if your netid is **asterix** and you are enrolled COMS 4720, then your submission file should be named **asterix-4720-hw2.pdf**. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed). Also, recall the AI policy:

- You may **not use AI tools to generate complete solutions** to assignments, coding problems, or any assessed work that is intended to measure your independent knowledge and skills.
- Whenever you use AI tools to assist in your assignment, you must **include a paragraph at the end of the assignment explaining what you used the AI for and include all the prompts you used to get the results**. Failure to do so is in violation of the academic honesty policies.

1. (20 pts) Which of the following are true and which are false? Explain your answers

- (a) (5 pts) Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.

False. DFS can be more efficient when the solution is on the first path.

- (b) (5 pts) $h(n) = 0$ is an admissible heuristic for the 8-puzzle.

True. $h(n) = 0$ will never overestimate the cost.

- (c) (5 pts) Breadth-first search is complete even if zero step costs are allowed

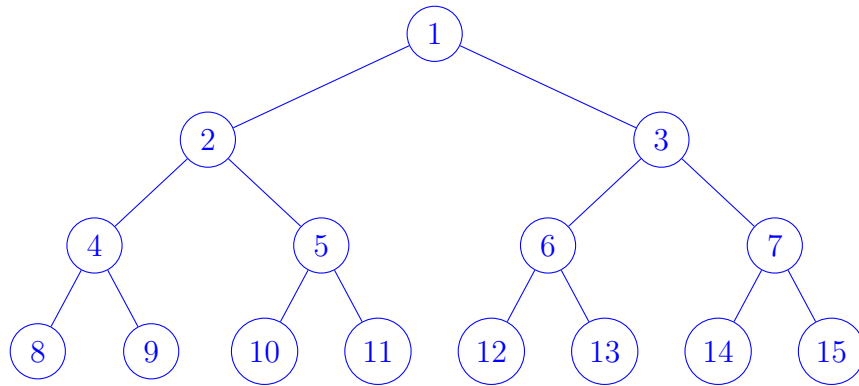
True. BFS is always complete when the branching factor is finite. Cost will not effect the completeness of BFS.

- (d) (5 pts) Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves.

False. In this problem, Manhattan distance can overestimate the cost since the rook can move more than one square each time.

2. (10 pts) Consider a state space where the start state is number 1 and each state k has two successors: numbers $2k$ and $2k + 1$.

(a) (5 pts) Draw the portion of the state space for states 1 to 15.



- (b) (5 pts) Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search (by depth layers).

Breadth-first search: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

Depth-limited search: 1, 2, 4, 8, 9, 5, 10, 11

Iterative deepening search:

Depth 0: 1;

Depth 1: 1, 2, 3;

Depth 2: 1, 2, 4, 5, 3, 6, 7;

Depth 3: 1, 2, 4, 8, 9, 5, 10, 11.

3. **(10 pts)** Describe a state space in which iterative deepening search performs much worse than depth-first search

In the state space which has n states and each state only has one successor (branching factor is 1), the DFS's time complexity is $O(n)$, while the iterative-deepening search's time complexity is $O(n^2)$.